Now that we have understood the mechanism for real numbers, what about $\mathbb{R}^2$? What about points in a Euclidean bidimensional space? What about $\mathbb{C}$ (the set of complex numbers).

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Notice that $\mathbb{R}^2$ and $\mathbb{C}$ are bidimensional Euclidean spaces, and entries of these spaces can be represented with a pair of real numbers.

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Well, if we move to higher dimensions, how we can run binary search is not as clear as the unidimensional case.

Binary search relies on recursively dividing a search space in halves, but if you consider a point P in the Cartesian plane, how can we partition the plane in two regions, left and right of P? Figure 8.8 shows a few possible ways to do so.

A first, visually intuitive approach could be splitting the plane along a vertical line passing through P, so that in our representation of the Cartesian plane, the two semi-spaces will actually be drawn on the left and on the right of P.